30.2 The Program class: Building a Web Host

All ASP.NET Core apps are fundamentally console applications. With the `Startup`-based hosting model, the `Main` entry point builds and runs an `IHost` instance, as shown in the following listing, which shows a typical Program.cs file. The `IHost` is the core of your ASP.NET Core application: it contains the HTTP server (Kestrel) for handling requests, along with all the necessary services and configuration to generate responses.

Listing 30.1 The Program.cs file configures and runs an `IHost`

```
public class Program
{
    public static void Main(string[] args)
    {
        CreateHostBuilder(args)                                    ❶
            .Build()                                               ❷
            .Run();                                                ❸
    }

    public static IHostBuilder CreateHostBuilder(string[] args) =>
        Host.CreateDefaultBuilder(args)                            ❹
            .ConfigureWebHostDefaults(webBuilder =>                ❺
            {
                webBuilder.UseStartup<Startup>();                  ❻
            });
}
```

❶ Creates an IHostBuilder using the CreateHostBuilder method

❷ Builds and returns an instance of IHost from the IHostBuilder

❸ Runs the IHost and starts listening for requests and generating responses

❹ Creates an IHostBuilder using the default configuration

❺ Configures the application to use Kestrel and listen to HTTP requests

❻ The Startup class defines most of your application's configuration.